



Reliability analysis and comparison of ring-PUF based on probabilistic models[☆]

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ABSTRACT

—To address the issue of maladaptive reliability theory models in delay-based physical unclonable functions (PUF) of ring classes, this paper begins by analyzing the relationship between four key transistor parameters (channel length, channel width, threshold voltage, and gate oxide thickness) and delay sensitivity. Then establishes reliability quantification formulas for various PUF types, including ring oscillator PUF, k-out-of-1 PUF, Loop PUF, transient effect ring oscillator PUF, and duty cycle PUF. This analysis and comparison are conducted using the inverter delay model and probabilistic statistical methods to ensure fairness at the theoretical level. Finally, the proposed model is validated through simulation experiments, along with further analysis and comparison of ring-PUF. The results demonstrate that delay sensitivity can be utilized to assess the extent of delay impact when process parameters fluctuate. In terms of reliability, the ring-PUF can be ranked as follows: k-out-of-1 PUF, Loop PUF, ring oscillator PUF, transient effect ring oscillator PUF, and duty cycle PUF. This paper offers extensive technical guidance and theoretical support for ring PUF designers.

1. Introduction

The advancement of information technology has posed significant challenges to the field of security. Digital keys, which are crucial for securing information, are typically stored in the non-volatile memory (NVM) of chips. However, there is a vulnerability in NVM that allows for tampering and unauthorized duplication of cryptographic data through reverse engineering and other methods [1]. Consequently, hardware security has emerged as a critical concern that researchers need to tackle.

Unlike traditional storage mechanisms in NVM devices, physically unclonable function (PUF) circuits generate unique digital information by leveraging the inherent random process variation in the chip manufacturing process [2]. PUF entropy sources typically employ symmetrical circuit structures that extract electrical characteristics such as voltage, current, or delay. These characteristics are then processed and compared to generate the response. PUFs act as secure encrypted

one-way functions, where a specified input challenge produces a unique output response. The mapping between the challenge and response is unpredictable, making PUFs unclonable and low-cost hardware security primitive that is resistant to physical attacks [3]. The hardware implementation of post-quantum cryptography (PQC) algorithms is susceptible to fault injection attacks, and the intrinsic PUF with low resource overhead is used to provide identification for PQC chips [4]. Emerging nanoelectronic devices are also widely used as PUF applications, such as the compatibility between resistive random access memory PUF and memory architecture, which can be used to reduce hardware overhead [5]. The reconfigurable PUF based on spin-transfer torque magnetoresistive random-access memory (STT-MRAM) significantly improves the flexibility and reliability of the PUF by implementing a cross-switching structure using a basic multiplexer [6].

PUFs have garnered significant attention in academia, leading to the development of various classical circuit designs such as the ring-oscillator PUF (RO PUF) [7], the arbiter PUF [8], and the SRAM PUF

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[9]. Generally speaking, if the number of CRPs supported by a PUF grows exponentially with its size, it is considered a strong PUF (Arbiter PUF), while linear or polynomial growth usually corresponds to a weak PUF (RO PUF, SRAM PUF). According to different application scenarios and different security level requirements, strong PUF and weak PUF play an irreplaceable role. Weak PUFs are often used for key generation in key encryption primitives and therefore must be lightweight and highly reliable. Strong PUF is often used in device authentication, requiring it to generate a huge space for CRPs, and it has a certain degree of fault tolerance for single-bit instability.

RO PUF are one of the most popular weak PUF architectures in the literature because they are easy to route and can ensure symmetric configurations on almost all FPGA platforms. RO PUF was proposed more than ten years ago, and the topic has been extensively studied since then. RO PUF research has been at the forefront of hardware security, both initially and now. Many methods, such as redundancy, configurable, enhanced measurement precision, comparison strategies, transistor-level optimization, and mathematical analysis, have been proposed to overcome the inherent problems and improve performance. Such as, a frequency offset-based RO PUF structure is proposed, which improves reliability by offsetting the frequency of RO pairs, resulting in a frequency difference greater than a given threshold [10]. A PRO PUF using programmable delay units is proposed, and the programmable interconnect points of programmable delay units are used to enhance reconfigurability [11]. A dual-state analog PUF (DA PUF) has been proposed, which can be flexibly configured as a low-state leakage PUF or a high-state inverter PUF to achieve lower temperature and voltage sensitivity [12]. At the same time, many novel ring structures have also been proposed, such as Loop PUF, Duty Cycle PUF and transient effect ring-oscillator PUF.

The RO PUF generates a one-bit response by comparing the oscillation frequencies of two symmetric rings. The outcome depends on the fluctuation of circuit delay caused by process variation during chip manufacturing. While the random and uncontrolled nature of process variation enhances the security of the RO PUF, it can also negatively impact circuit reliability, particularly with small process variations. The response of the RO PUF can be affected by unstable flip-flops caused by noise, ambient temperature, and voltage fluctuations when the delay fluctuations due to process variation are minimal [13]. Researchers have been investigating and attempting to address this reliability issue.

Suh et al. [7] identified two key factors that affect the reliability of the RO PUF. Firstly, the output response is highly sensitive to temperature, meaning that the output tends to change with increasing temperature. Secondly, if a ring is involved in multiple comparisons, this can lead to correlations between responses. To address these issues, a 1-out-of- k shielding mechanism was proposed, which screens out the largest period difference from the k rings to generate a response. This approach significantly improves the robustness of the response to temperature. However, it is important to note that this redundant and fault-tolerant approach results in an exponential hardware overhead. Maiti et al. [14] introduced a configurable RO PUF structure. In this structure, a multiplexer is included after each stage of the inverter in the RO. Through the multiplexer, different heterogeneous rings are implemented, and the response in the same group of rings is generated by filtering the one with the largest delay difference. This configurable structure helps in reducing the overall entropy of the circuit and saving hardware resources. However, it results in lower reliability compared to the 1-out-of- k shielding mechanism.

Agustin et al. [15] proposed a Duty-Cycle PUF (DC PUF) structure that measures the duty cycle of a ring oscillator (RO) as a response. They demonstrated that the duty cycle is more robust to temperature and voltage variations. However, this method is insensitive to process variation and faces difficulty in accumulating sufficient delay fluctuations in mature manufacturing processes. Danger et al. [16,17] proposed a Loop PUF structure consisting of N identical and controllable delay chains of ROs. Each chain contains M configurable cells and produces a response

by comparing the frequencies of the two ROs.

Bossuet et al. [18] designed a transient effect ring-oscillator PUF (TERO PUF) with an even number of inverters. It generates a response by comparing the number of oscillations and has a structure that exhibits good resistance to logic locking and electromagnetic interference. Herkle et al. [19] discovered that counter failure significantly affects the reliability of the RO PUF. They proposed a feedback RO that effectively shields glitches. However, no analysis of the oscillation behavior or proof of stability for this circuit was provided.

To comprehensively evaluate the performance of ring PUFs, Wild et al. [20] conducted a study using 100 FPGA devices. They implemented RO PUF, Loop PUF, and TERO PUF and found that RO PUFs exhibited the highest reliability in temperature varying environments. Similarly, Zulfikar et al. [21] also used FPGA devices and concluded that the length of the RO PUF had a positive correlation with its reliability. These findings indicate that the reliability of PUFs is heavily influenced by the platform on which they are implemented. However, it is important to note that the results obtained from a limited number of FPGA devices may not be representative of a general pattern. On a different note, Maes et al. [22] developed a probabilistic model to assess the reliability of SRAM PUFs, which offers a theoretical foundation for studying the reliability of storage PUFs. In their study, Lao et al. [23] developed a probabilistic statistical model that analyzed the reliability of several arbiter PUFs based on the delay of a multiplexer. Similarly, Danger et al. [24] developed a prediction model that quantified the relationship between noise level and delay class PUF reliability, as well as entropy, using the signal-to-noise ratio. Building upon these works, Xu et al. [25] incorporated transistor-level analysis into the delay-based probabilistic statistical model to perform reliability analysis of arbiter PUFs, RO PUFs, and DC PUFs. They utilized the channel width of the transistor as a source of process variation. However, the authors did not specifically explore the impact of different configurations of process parameters on PUF reliability or the extent to which different process parameters affect PUF reliability.

To address the reliability challenges encountered by delay-based ring PUFs and the imperfections in transistor-level theoretical models, this paper examines the correlation between four key process parameters: channel length, channel width, threshold voltage, and gate oxide thickness, and the delay sensitivity of transistors. Additionally, the paper quantitatively models the reliability of various commonly used ring PUFs. Simulation experiments are conducted to verify the accuracy of the proposed models and to perform a fair comparison of ring PUFs.

2. Preliminary

A. Process Variations

Variations in transistor process parameters during fabrication play a crucial role in influencing circuit performance. As feature sizes continue to shrink, process variation becomes a more significant obstacle in the design of reliable integrated circuits. In most design and application scenarios, minimizing process variation leads to improved circuit performance [26]. However, hardware security primitives like PUFs require a certain level of process variation to ensure their security properties. Understanding the sources of process variation and their impact on the electrical characteristics of the circuit is therefore essential in PUF design.

The manufacturing process of integrated circuits is divided into two main stages: the front-end-on-line and the back-end-on-line. The front-end process involves producing transistors and active devices on the surface of the silicon wafer, while the back-end process focuses on implementing interconnects and passive devices. The silicon wafers undergo various complex processes, including grinding, polishing, lithography, etching, doping, and deposition. However, during the photolithography and etching stages, certain issues such as line edge roughness and optical proximity effects can lead to fluctuations in the

transistor's channel length and width. Additionally, ensuring a stable thickness for the gate oxide layer during the growth process is also challenging. The final threshold voltage is influenced by the thickness of the electrolyte layer, carrier concentration, and the flatness of the doping profile. As depicted in Fig. 1, process variations result in fluctuations in the physical parameters of the transistor, including channel length, channel width, threshold voltage, and gate oxide thickness. These variations impact the transistor's I-V characteristics and ultimately cause changes in performance, such as the delay of the logic gate.

B. Inverter Delay Models

Sakurai et al. [27] proposed the alpha power law model to simplify the full transistor model by considering the leakage current. They established a quantized formula (1) with factors such as load capacitance, threshold voltage, channel length, channel width, gate oxide thickness, carrier's mobility, and dielectric constant of the transistor.

$$t_{inv} = \frac{C_l V_{dd} L T_{ox}}{\mu \epsilon_{ox} W (V_{dd} - V_{th})^\alpha} \quad (1)$$

where C_l refers to the load capacitance at the output of the inverter, V_{dd} refers to the supply voltage, L and W refer to the channel length and width of the transistor respectively, T_{ox} refers to the gate oxide thickness, μ refers to the carrier's mobility, ϵ_{ox} refers to the gate oxide permittivity, V_{th} refers to the threshold voltage and α refers to the Sakurai constant.

3. Delayed sensitivity analysis

From formula (1), it can be observed that there exists a specific functional relationship between the process parameters and the delay of the inverter. In order to quantify the extent of influence on the delay caused by fluctuations in the process parameters, the delay sensitivity (D_s) is introduced. The delay sensitivity for a particular parameter can be obtained by taking the first-order derivative of formula (1). Formulas (2) - (5) represent the delay sensitivities for channel length, channel width, gate oxide thickness, and threshold voltage, respectively.

$$D_{s-L} = \frac{\partial t}{\partial L} = \frac{C_l V_{dd} T_{ox}}{\mu \epsilon_{ox} W (V_{dd} - V_{th})^\alpha} \quad (2)$$

$$D_{s-W} = \frac{\partial t}{\partial W} = - \frac{C_l V_{dd} L T_{ox}}{\mu \epsilon_{ox} W^2 (V_{dd} - V_{th})^\alpha} \quad (3)$$

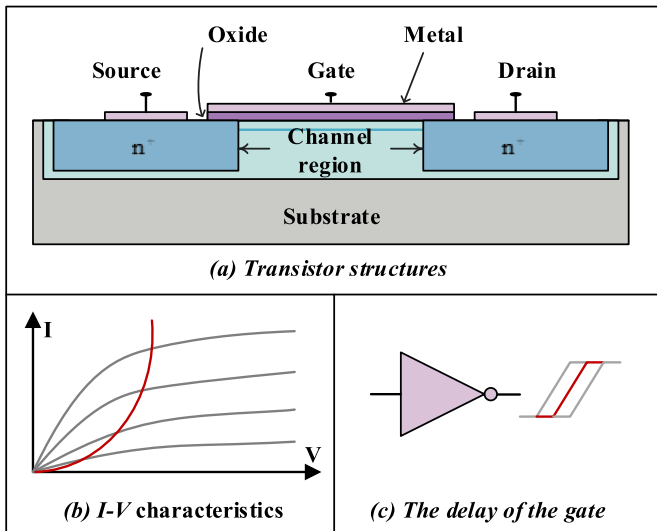


Fig. 1. The relationships between process variations and the performance of the device.

$$D_{s-T_{ox}} = \frac{\partial t}{\partial T_{ox}} = \frac{C_l V_{dd} L}{\mu \epsilon_{ox} W (V_{dd} - V_{th})^\alpha} \quad (4)$$

$$D_{s-V_{th}} = \frac{\partial t}{\partial V_{th}} = - \frac{\alpha C_l V_{dd} L T_{ox}}{\mu \epsilon_{ox} W (V_{dd} - V_{th})^{\alpha+1}} \quad (5)$$

According to formulas (2) - (5), it can be observed that the channel length and delay sensitivity are directly proportional. This means that as the channel length increases, the impact of process variation on the inverter delay becomes more significant. On the other hand, the channel width and delay sensitivity are inversely proportional. This implies that as the channel width increases, the inverter delay is less affected by process variation. Additionally, there is a positive relationship between the gate oxide thickness and threshold voltage with delay sensitivity.

The manufacturing process variation of L , W , T_{ox} , and V_{th} is assumed to obey normal distribution, and correlations between various types of parameters are neglected. By applying the second moment (variance) to the expanded expression of the delay function and after some operations, a first-order expression of the variance of the gate delay is obtained [29,30]. The variance (σ^2) of the inverter delay for variations in parameters is obtained by expanding D_{s-L} , D_{s-W} , $D_{s-T_{ox}}$, and $D_{s-V_{th}}$ as follows:

$$\sigma^2 = (D_{s-L} \sigma_L)^2 + (D_{s-W} \sigma_W)^2 + (D_{s-T_{ox}} \sigma_{T_{ox}})^2 + (D_{s-V_{th}} \sigma_{V_{th}})^2 \quad (6)$$

In the design of PUF circuits, a high delay sensitivity is actually beneficial as it allows for efficient accumulation of process variation, which is essential for a stable PUF function. Therefore, the designer needs to consider the quality of the entropy source of the PUF circuit in terms of the cumulative efficiency of process variation while implementing its logic function. Generally, increasing the channel length, gate oxide thickness, and threshold voltage, while reducing the channel width, improves the reliability of the PUF, as long as the circuit operates correctly.

4. Ring-PUF reliability model

Based on the analysis of the inverter delay sensitivity, there is a strong correlation between the degree of inverter delay fluctuation and the process variation of key parameters. Process variation occurs randomly and is assumed to follow a normal distribution centered on a nominal value. Similarly, the delay fluctuations of the inverter can be considered to follow a normal distribution with certain nominal values, as shown by the following formula:

$$t_r \sim N(t_0, \sigma_r^2) \quad (7)$$

$$t_f \sim N(t_0, \sigma_f^2) \quad (8)$$

where t_r is the inverter rising time and t_f is the falling time. In general, $t_r = t_f = t_0$. The σ_r and σ_f indicate the standard deviation of the distribution and reflect the concentration of delay fluctuations. Generally, there is $\sigma_r = \sigma_f = \sigma_0$, the magnitude of which depends on the channel length, channel width, gate oxide thickness and the delay sensitivity of the threshold voltage and the level of process maturity.

A. Ring Oscillator PUF

Ring Oscillators are crucial in integrated circuit design and find extensive applications in pulse-width modulation, phase-locked loops, and time-to-digital conversion. They are composed of a series of first- and tail-linked inverters, with an odd number of stages, which produce an oscillation signal with a 50 % duty cycle. RO PUFs utilize ROs as the primary source of entropy [7] and are typically organized as illustrated in Fig. 2. A multiplexer is employed to select two ROs (RO1, RO2) based on the given challenge and connects them to a counter. The counter measures the number of oscillations during a specific sampling time,

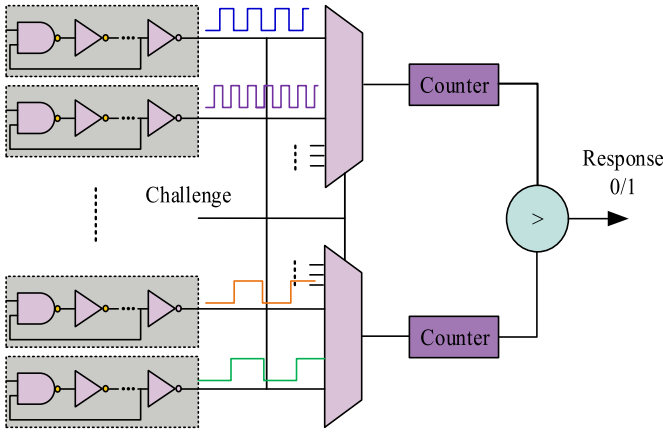


Fig. 2. The overall structure of the RO PUF.

thus determining the period of the two ROs. The design principle of the RO PUF ensures that all the ROs have identical layouts and wiring, which theoretically guarantees equal periods. Consequently, the final response solely relies on process variation during the manufacturing stage.

In the common RO PUF, the strategy for selecting ROs involves generating a one-bit response whenever two adjacent ROs are chosen, without reusing any RO. This means that N ROs can produce $N/2$ independent bit responses. Let's assume that each RO contains M inverters, and the correlation coefficient between the rising and falling times of the inverters is denoted as ρ_a , where $\rho_a \in [0, 1]$. According to formula (1), the rising and falling times are usually positively correlated with the influence of process variations, so ρ_a must be greater than 0. The period T of the RO can be calculated as $2Mt_0$. Formula (9) is derived from the properties of the normal distribution.

$$T \sim N\left(2Mt_0, M\left(\sigma_r^2 + \sigma_f^2 + 2\rho_a\sigma_r\sigma_f\right)\right) \\ = N\left(2Mt_0, 2M(1 + \rho_a)\sigma_0^2\right) \quad (9)$$

Similarly, the period difference ΔT between the two RO follows a normal distribution of:

$$\Delta T \sim N(0, 4M(1 + \rho_a)\sigma_0^2) \quad (10)$$

When the difference (ΔT) is greater than the threshold value (w), the output response is 1. Conversely, when ΔT is less than or equal to $-w$, the

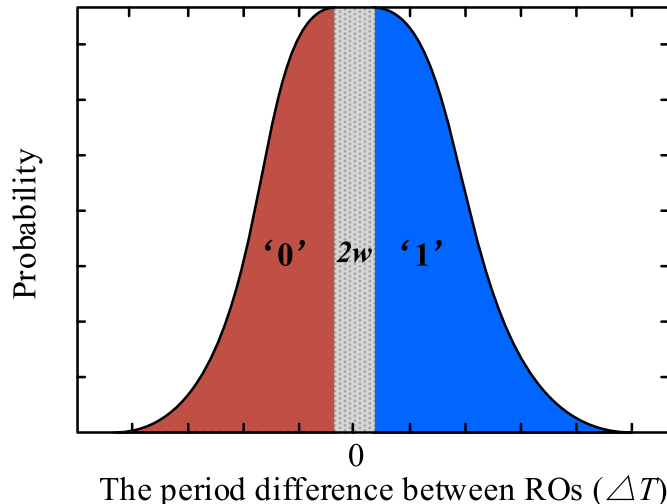


Fig. 3. The response distribution of RO PUF.

output response is 0. In Fig. 3, the probability distribution is illustrated, with the red area representing a response of 0 and the blue area representing a response of 1. The grey area indicates an unstable result, where a slight variation in ΔT can cause the response to fluctuate due to factors such as noise, temperature variations, voltage fluctuations, and counter accuracy. Therefore, minimizing the size of the grey area enhances the reliability of the PUF.

Based on the above analysis, RO PUF reliability is equal to the cumulative distribution probability that $|\Delta T|$ is greater than w , i.e:

$$R_{ro-puf} = P(|\Delta T| > w) \\ = 1 - \int_{-w}^w \frac{1}{\sqrt{2\pi} \sqrt{4M(1 + \rho_a)\sigma_0^2}} \exp\left(-\frac{x^2}{2 \times 4M(1 + \rho_a)\sigma_0^2}\right) dx \\ = 1 - \text{erf}\left(\frac{w}{2\sqrt{2M(1 + \rho_a)\sigma_0}}\right) \quad (11)$$

The reliability of the 1-out-of- k mechanism can be increased by using redundancy, as shown in Ref. [7]. In this mechanism, the 8 ROs are divided into 4 pairs, and the pair with the largest period difference is selected. The reliability under this mechanism can be calculated as:

$$R_{1-out-of-8} = 1 - \left[\text{erf}\left(\frac{w}{2\sqrt{2M(1 + \rho_a)\sigma_0}}\right)\right]^4 \quad (12)$$

B. Loop PUF

The structure of the loop PUF is depicted in Fig. 4. It consists of a series of N delay chains, forming a closed odd-order ring oscillator. The delay chains are connected by inverters and AND gate, with the control module providing the input to the AND gate to control the start and stop of oscillation. Each delay chain comprises M controlled delay elements connected to each other, where each delay element can be independently controlled to choose the top or bottom path. External challenges are transmitted to each delay chain through the control module, allowing the selection of different paths for the target delay unit. Subsequently, the control module sends a signal to the AND gate to initiate the oscillation of the entire loop. The counter measures the oscillation period. Finally, compare the two measured periods to obtain the response results.

Obviously, the Loop PUF also measures and compares the oscillation period of the ROs as the response. In comparison to the RO PUF, the Loop PUF sequentially compares multiple delay chains, offer greater flexibility and ease of design. The comparison strategy employed by the Loop PUF has a significant impact on the reliability of the response, and this paper adopts the same approach as the RO PUF. In each selected delay chain, all cells are chosen as 1, while the remaining cells in the chain are chosen as 0. That is to only consider the comparison between ROs composed of absolutely independent delay units. Therefore, similar to the RO PUF, the reliability of the single-bit response from the Loop PUF can be determined using formula (13):

$$R_{loop-puf} = 1 - \text{erf}\left(\frac{w}{2\sqrt{2M(1 + \rho_a)\sigma_0}}\right) \quad (13)$$

C. Duty cycle PUF

Further research reveals that the stability of the response in RO PUF is closely connected to variations in voltage and temperature. Robustness to PVT is better demonstrated by the duty cycle, and a novel approach proposed in Ref. [15] suggests measuring the duty cycle instead of the period. To enhance the number of challenge-response pairs, an asymmetric inverter structure is designed to accommodate varying duty cycles for the nodes within the RO. The primary structure

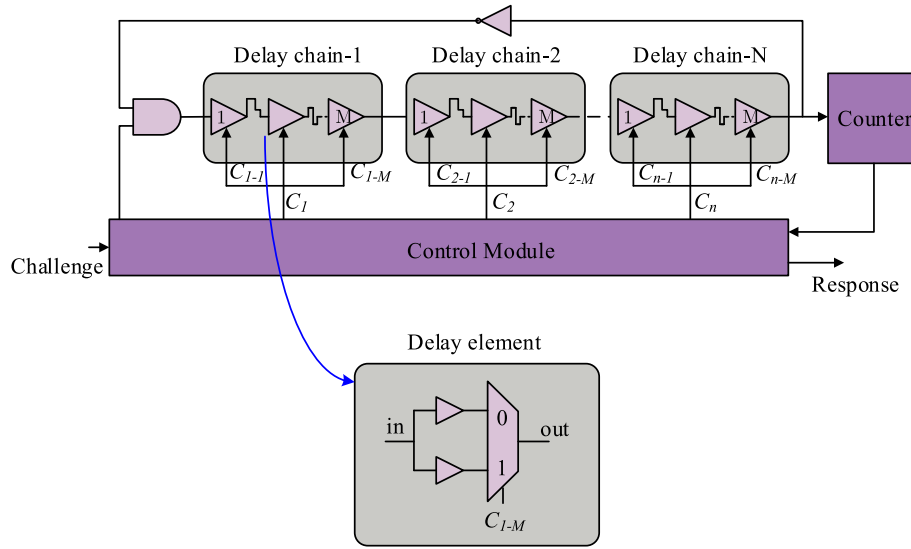


Fig. 4. The overall structure of Loop PUF.

of the DC PUF is illustrated in Fig. 5, and it comprises N identical ROs as its main components. Challenge-(y) is implemented to designate the output node of the RO, while challenge-(x) is employed to select the specific RO. Counter-Tlow1 utilizes a high-speed signal to sample the output of the first selected RO for obtaining the low-level time, whereas Counter-T1 calculates the period of the low-level time. The duty cycle value is determined by dividing the two aforementioned results. Counter-Tlow2 and Counter-T2, on the other hand, are responsible for measuring the duty cycle of the second RO. Ultimately, the obtained values are compared to determine whether the output response is 1 or 0.

For an RO consisting of M inverters, the high-level time (t_h) and low-level time (t_l) of the period signal can be expressed as formulas (14) and (15), respectively. Here, $t_{r,M}$ represents the rise time of the M -th inverter and $t_{f,M}$ represents its fall time. Consequently, the duty cycle (D_{RO}) can be expressed using formula (16):

$$t_h = t_{f,M} + \sum_{i=1}^{(M-1)/2} (t_{r,2i} + t_{f,(2i-1)}) \quad (14)$$

$$t_l = t_{r,M} + \sum_{i=1}^{(M-1)/2} (t_{f,2i} + t_{r,(2i-1)}) \quad (15)$$

$$D_{RO} = \frac{t_l}{t_h + t_l} = \frac{t_{r,M} + \sum_{i=1}^{(M-1)/2} (t_{f,2i} + t_{r,(2i-1)})}{\sum_{i=1}^M (t_{f,i} + t_{r,i})} \quad (16)$$

For the sake of simplicity and fairness, this paper focuses on only one node of RO and does not consider the asymmetric design. Formula (16) shows a strong correlation between the numerator and denominator. Assuming the correlation coefficient between the rising times and falling times in the numerator and denominator in formula (16) is ρ_d . Since the weights of the numerator and denominator are equal, the operation of the quotient will weaken the impact of process variations. So, $\rho_d \in [-1, 0]$. The duty cycle D_{RO} approximately follows the distribution mentioned below:

$$D_{RO} \sim N\left(\frac{1}{2}, M(1 + \rho_d)\sigma_0^2\right) \quad (17)$$

Then the difference between the two RO duty cycles, ΔD_{RO} , follows the following distribution:

$$\Delta D_{RO} \sim N(0, 2M(1 + \rho_d)\sigma_0^2) \quad (18)$$

Referring to formula (11), the reliability of the DC PUF can be derived as:

$$R_{dc-puf} = 1 - \text{erf}\left(\frac{w}{2\sqrt{M(1 + \rho_d)}\sigma_0}\right) \quad (19)$$

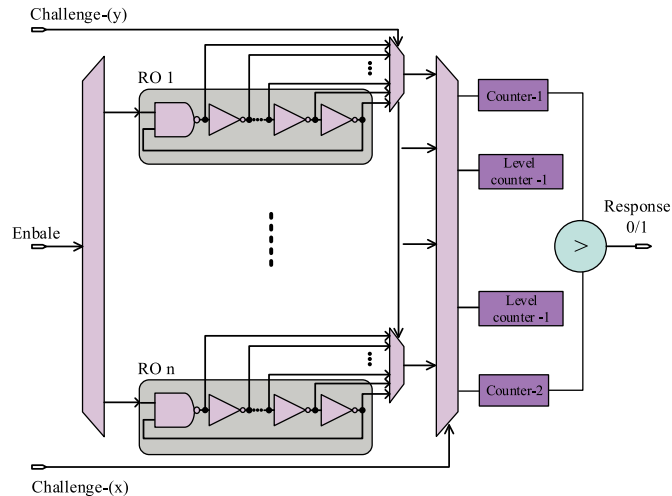


Fig. 5. The overall structure of DC PUF.

D. Transient Effect Ring Oscillator PUF

The TERO PUF is a modified version of the RO PUF where the RO is replaced with TERO. TERO can be understood as an SR flip-flop with a symmetrical chain of inverters added after the two NAND gates, as depicted in Fig. 6.

After the input controls of both NAND gates are pulled high simultaneously, the signals S1 and S2 follow different paths. As there are an even number of inverters, each signal is flipped in the same direction on each inverter. Specifically, S1 changes from 0 to 1 on odd-numbered inverters and from 1 to 0 on even-numbered levels, while S2 behaves in the opposite manner. Due to process variation, there is a delay difference between S1 and S2, causing the signals to eventually collide and disappear after a certain number of oscillations. The number of oscillations is directly related to the process variation. The theoretical number of oscillations C can be calculated from the delay of S1 and S2, T being the theoretical delay of the TERO ring for one loop. Then we have:

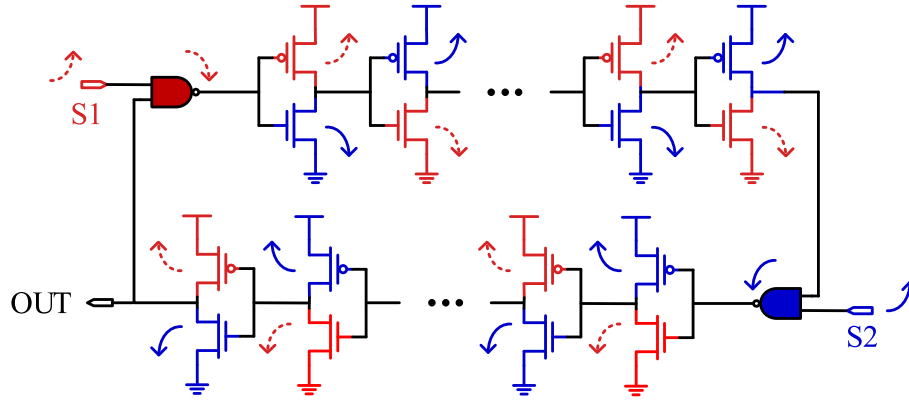


Fig. 6. The circuit of the transient effect ring oscillator.

$$t_{S1} = \sum_1^{(M-1)/2} (t_{r,2i} + t_{f,(2i-1)}) \quad (20)$$

$$t_{S2} = \sum_1^{(M-1)/2} (t_{f,2i} + t_{r,(2i-1)}) \quad (21)$$

$$C = \frac{T}{|t_{S1} - t_{S2}|} \quad (22)$$

Assuming that the correlation coefficient between the rising times and falling times in the numerator and denominator in formula (22) is denoted as ρ_s . It is important to note that the numerator involves a sum operation, while the denominator involves a subtraction operation. As a consequence, the weight of the numerator is greater than that of the denominator, resulting in a partial weakening of the impact of the final process variation. Therefore, it can be concluded that $\rho_s \in [-1, 0]$, with ρ_s being greater than ρ_d . The number of oscillations (C) can be approximated by the following distribution:

$$C \sim N(\infty, 2M(1 + \rho_s)\sigma_0^2) \quad (23)$$

Similarly, the difference ΔC between the number of oscillations of the two TEROs follows a normal distribution as described below:

$$\Delta C \sim N(0, 4M(1 + \rho_s)\sigma_0^2) \quad (24)$$

Referring to formula (11), the reliability of the TERO PUF can be derived as:

$$R_{tero-puf} = 1 - \text{erf}\left(\frac{w}{2\sqrt{2M(1 + \rho_s)\sigma_0^2}}\right) \quad (25)$$

5. Experimental results

To verify the correctness of the above model and to perform a comparison of the reliability of ring PUFs, this section uses the Hspice tool to build a simulation environment for the relevant circuits based on the PTM (Predictive Technology Model) 45 nm process library [28]. The Monte Carlo method is used to simulate the random process variation generated during the manufacturing process. The sources of process variation are set as the channel length, channel width, gate oxide thickness and threshold voltage of the transistor.

A. Delay Sensitivity

Delay sensitivity is a measure of how changes in process parameters affect the delay. In this experiment, we focus on the delay fluctuations of the inverter when there are random variations in the four main process parameters. We set the process parameters to follow a 3σ normal distribution within $\pm 10\%$ of the calibration value in a Monte Carlo simulation. The standard inverter parameters used are as follows: voltage - 1.1 V, channel length - 45 nm, channel width - 121.5 nm, gate

oxide thickness - 1.0 nm, carrier concentration - $5 \times 10^{14} \text{ nm}^2/\text{Vs}$, gate oxide dielectric constant - $3.9 \times 8.85 \times 10^{-21} \text{ F/nm}$, threshold voltage - 0.412 V, and Sakurai constant - 1.5.

The delay sensitivity of the inverter was calculated for different channel lengths (45 nm–90 nm) using formula (2). The standard deviation of the delay fluctuations for the corresponding channel length variation was obtained through Hspice Monte Carlo simulations. The results are shown in Fig. 7. It was observed that the delay sensitivity tends to increase as the channel length increases (17 ps/nm - 69 ps/nm). Similarly, the standard deviation obtained from the simulation also increases (0.85 ps - 36.502 ps).

The delay sensitivity for channel width (90 nm–180 nm), gate oxide thickness (1 nm–2 nm), and threshold voltage (0.233 V–0.699 V) is also calculated using formulas (3) - (5) and compared with the delay standard deviation obtained from Hspice simulation. In Fig. 8, it can be observed that as the channel width increases, the delay sensitivity gradually decreases from 17.454 ps/nm to 17.416 ps/nm. Similarly, the standard deviation obtained from the simulation decreases from 0.2 ps to 0.156 ps. The delay sensitivity increases from 17.454 ps/nm to 28.097 ps/nm with an increase in gate oxide thickness, as shown in Fig. 9. Similarly, Fig. 10 demonstrates that the delay sensitivity increases from 11.041 ps/nm to 34.663 ps/nm with an increase in threshold voltage. Furthermore, the simulation results indicate that the standard deviation increases from 0.066 ps to 0.676 ps.

Based on the results, it can be concluded that: a. delay sensitivity characterizes the extent to which process variation affects delay; b. delay fluctuations are highly influenced by channel length and show a

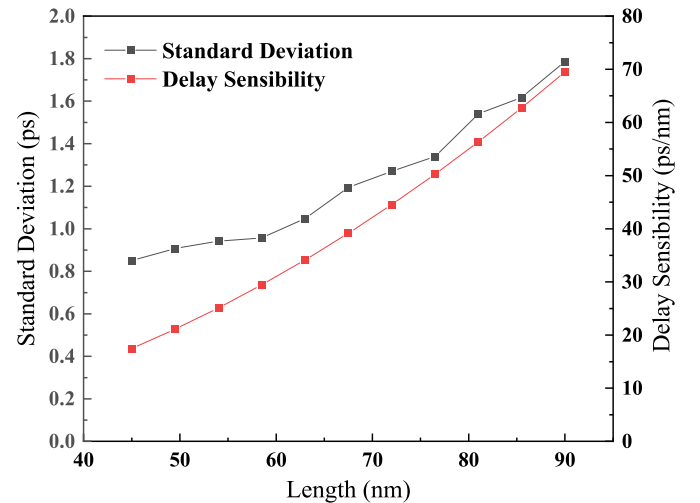


Fig. 7. The results of delay sensitivity in the model and delay standard deviation in simulation – channel length.

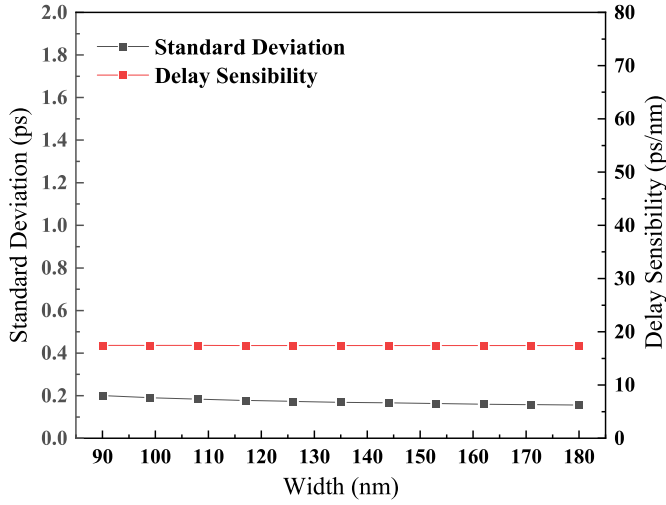


Fig. 8. The results of delay sensitivity in the model and delay standard deviation in simulation – channel width.

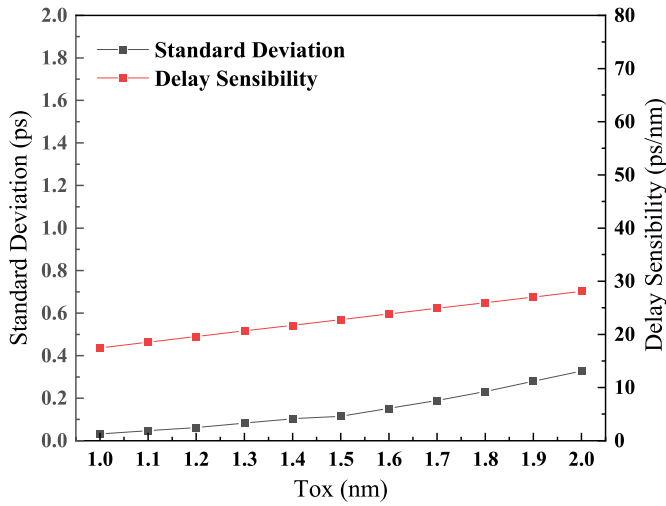


Fig. 9. The results of delay sensitivity in the model and delay standard deviation in simulation – the gate oxide thickness.

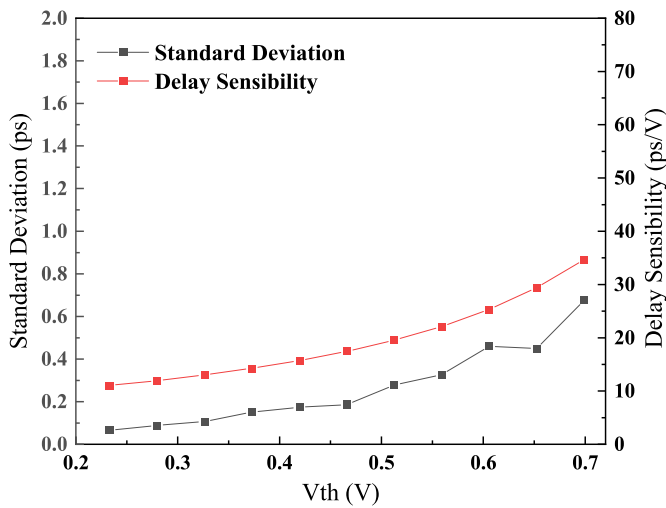


Fig. 10. The results of delay sensitivity in the model and delay standard deviation in simulation –the threshold voltage.

positive correlation; c. gate oxide thickness and threshold voltage have a weaker impact on delay compared to channel length, but also exhibit a positive correlation; d. delay fluctuations are least affected by channel width and demonstrate a negative correlation.

B. Reliability Comparison

To validate the ring PUF reliability models developed in this paper as well as to fairly compare their advantages and disadvantages. In this section, the reliability of various ring PUFs (where ρ_a is 0.2, ρ_d is -0.3 , ρ_s is -0.1 , w is 1 ps and σ_θ is 1 ps) is calculated by formulas (11), (12), (13), (19) and (25). The above parameter values are all derived from theoretical and empirical results of SPICE simulation. Meanwhile, the above ring PUFs were implemented using 4096 rings respectively, and the channel length, channel width, gate oxide thickness and threshold voltage were set as the sources of process variation in Hspice's Monte Carlo simulation, and the experimental results are shown in Figs. 11 and 12.

When comparing Figs. 11 and 12, it is evident that the model results presented in this paper align well with the experimental results obtained through simulation. The reliability of the ring PUFs shows an upward trend as the length of the ring increases. Notably, the 8-out-of-1 mechanism demonstrates exceptional reliability, achieving a reliability rate of 99.93 % even at smaller ring lengths. Conversely, the DC PUF exhibits the lowest reliability among all the structures. The comparison between the model results and simulation results for the DC PUF in Figs. 11 and 12 reveals a significant difference. The duty cycle measurement results in SPICE simulation exhibit notable resilience to process variations. The increase in the length of RO does not significantly improve the reliability of DC PUF. This suggests that the simplification of the correlation coefficient in formula (19) is overly optimistic and does not accurately reflect real-world scenarios. It is precisely for this reason that the DC PUF exhibits the lowest reliability among all PUFs.

Referring to formulas (10) and (11), it can be observed that both the RO PUF and the Loop PUF exhibit similar reliability. In the simulation experiments, it was observed that the Loop PUF exhibits slightly higher reliability compared to the RO PUF. This can be attributed to the longer interconnects in the Loop PUF, which increase the load on the inverter and consequently the delay sensitivity. As a result, the Loop PUF achieves slightly higher reliability even with the same process deviation. Among the ring PUFs, the reliability ranking from high to low is as follows: 8-out-of-1, 4-out-of-1, Loop PUF, RO PUF, TERO PUF, and DC

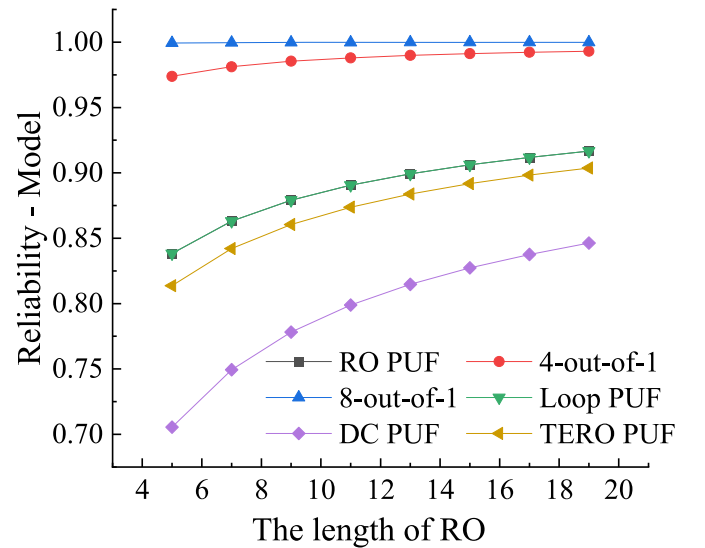


Fig. 11. The relationship between ring PUF reliability and length - Model results.

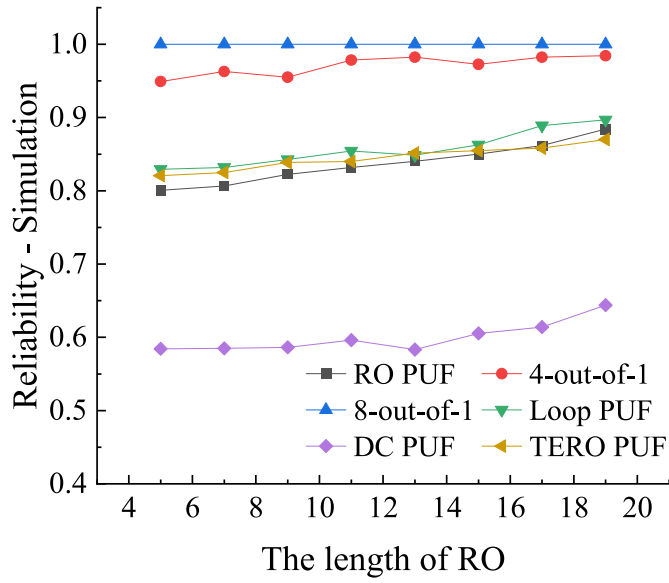


Fig. 12. The relationship between ring PUF reliability and length - Simulation results.

PUF.

To examine the impact of counter accuracy and noise level on the reliability of the PUF, we analyzed the results of model calculations and simulation experiments. The level of noise can be characterized by changing the boundary (w) in Fig. 3. After extensive analysis of experimental data, we have determined that a noise range of 0.5 ps–4.0 ps is appropriate based on our experience. Fig. 13 demonstrates the model result of the effect on reliability, and Fig. 14 shows the simulation result of the impact of noise level at a fixed ring length. The findings reveal that the reliability of the ring PUF declines significantly when the accuracy of the calculator and the noise level worsens. Notably, the 8-out-of-1 mechanism exhibits remarkable resistance to noise.

C. Comprehensive Analysis

This section provides a comprehensive comparison of the reliability of several common ring PUFs through model analysis and simulation experiments. The 8-out-of-1 mechanism relies on circuit redundancy

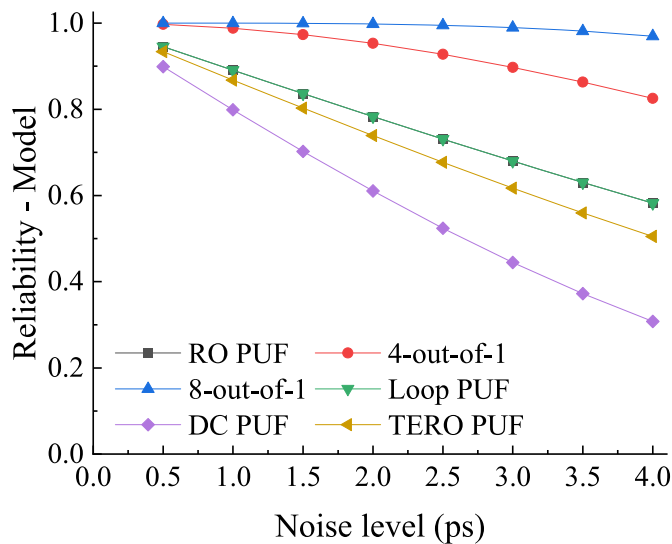


Fig. 13. The relationship between ring PUF reliability and noise level - Model results.

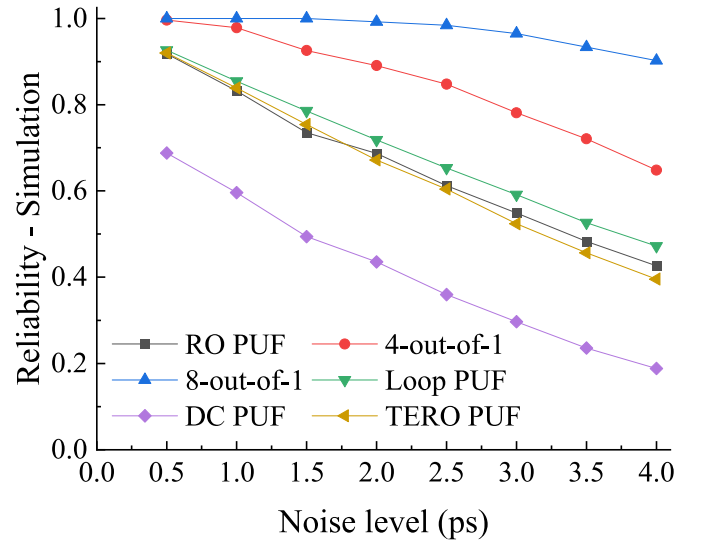


Fig. 14. The relationship between ring PUF reliability and noise level - Simulation results.

and fault tolerance to achieve better reliability results, but at the cost of hardware overhead. However, in circuit design, both hardware overhead and reliability are equally important considerations. Therefore, the selection of the best PUF solution should be based on the specific application requirements. Fig. 15 illustrates the relationship between hardware overhead and reliability for ring PUFs. Loop PUFs, RO PUFs, and TERO PUFs offer cost-effective options, while the k-out-of-1 mechanism allows for efficient utilization of hardware at the expense of reliability improvements.

6. Conclusion

This paper examines the impact of transistor parameters on the reliability of ring PUFs. The analysis focuses on delay sensitivity and concludes that increasing the channel length, gate oxide thickness, and threshold voltage, while reducing the channel width, can improve PUF reliability. Additionally, a quantitative reliability formula is developed for ring PUFs by combining the inverter delay model and the statistical probability method. The theoretical model is used to conduct a comprehensive comparison of the reliability of different ring PUFs.

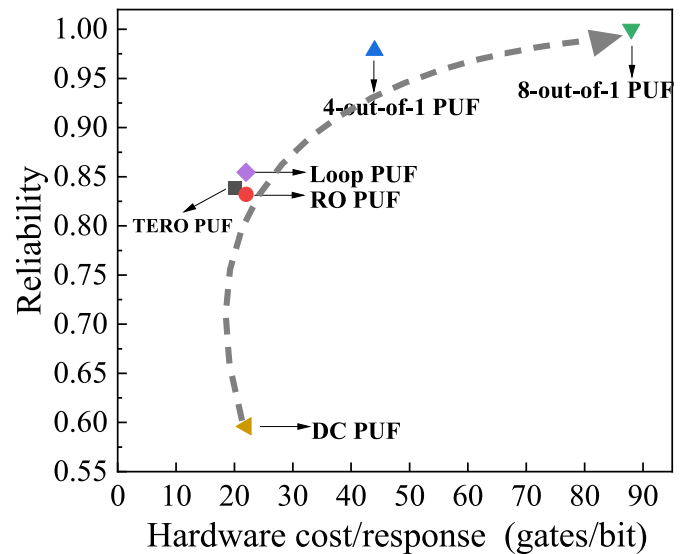


Fig. 15. The ring PUF area overhead and reliability trade-offs.

Simulation experiments are performed to verify the proposed model and further analyze and compare ring PUFs. The research approach to ring PUF reliability in this paper provides comprehensive technical guidance and theoretical support for designers.

Increasing the channel length, gate oxide thickness, and threshold voltage, while reducing the channel width, can enhance the reliability of the PUF, as long as the circuit operates correctly. Some simple design principles include selecting the smallest STD cell type, using low-power cells, employing long wires whenever feasible, and incorporating multiple stages of length.

In our future work, we plan to implement FPGA and utilize machine learning techniques to accurately predict the reliability and other performance parameters of various PUFs, even with limited data.

Authorship contributions

Please indicate the specific contributions made by each author (list the authors' initials followed by their surnames, e.g., Y.L. Cheung). The name of each author must appear at least once in each of the three categories below.

Conception and design of study: Jingchang Bian, Yankun Lin, Zhao Yang; Acquisition of data: Jingchang Bian, Zhengfeng Haung; Analysis and/or interpretation of data: Huaguo Liang, Tianming Ni. Drafting the manuscript: Jingchang Bian; Revising the manuscript critically for important intellectual content: Zhengfeng Haung. Approval of the version of the manuscript to be published (the names of all authors must be listed): Jingchang Bian, Zhengfeng Huang, Yankun Lin, Zhao Yang, Huaguo Liang and Tianming Ni.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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